

Complete Project Report

AI Student Helpdesk Assistant

Academic Project Report

Submitted in partial fulfillment of the requirements for the degree of
[Degree Name] in [Department Name]

Submitted By:

- [Student Name 1] (Roll No: [Roll Number 1])
- [Student Name 2] (Roll No: [Roll Number 2])

Under the Guidance of:

[Guide Name]

[Guide Designation]

[College Name]

[Academic Year]

DECLARATION

We hereby declare that the project entitled "AI Student Helpdesk Assistant" submitted to [College Name] is a record of an original work done by us under the guidance of [Guide Name].

ACKNOWLEDGEMENT

We express our sincere gratitude to [Guide Name] for their invaluable guidance.

ABSTRACT

The AI Student Helpdesk Assistant is a centralized academic management and self-service portal designed to streamline information retrieval in educational institutions. The system replaces manual enquiry processes with an intelligent Extractive QA Chatbot (built using TF-IDF and Cosine Similarity) that provides instant answers from college-uploaded documents (PDF/DOCX), alongside real-time updates on assignments, timetables, and announcements.

Table of Contents

1. Introduction
2. Existing and Proposed System
3. Requirement Analysis
4. System Architecture
5. Database Design

6. System Modules
7. AI Chatbot Implementation
8. API Documentation
9. Testing
10. Conclusion

Chapter 1: Introduction

1.1 Background

Educational institutions generate vast amounts of operational and academic information. Students often struggle to find answers to routine questions, leading to administrative overhead.

1.2 Problem Statement

Information is distributed across notice boards, disparate documents, and departments. Finding academic information is time-consuming, resulting in repeated queries to the administration.

1.3 Project Objectives

- Centralize academic information (timetables, assignments, announcements).
- Provide a Student Self-Service Portal.
- Implement an AI-based chatbot using NLP techniques (TF-IDF) to extract answers from uploaded PDFs and DOCX files.
- Deliver an Administrative Dashboard for real-time analytics.

1.4 Project Scope

The project covers Admin capabilities (managing students, subjects, materials, etc.) and Student capabilities (interacting with the AI, viewing timetables/assignments). It operates offline using a local database and ML libraries, strictly adhering to zero-internet-dependency rules.

1.5 Limitations

- The AI relies on Extractive QA (TF-IDF), so it extracts verbatim text rather than generating new contextual sentences (LLM).
- Scalability is bounded by SQLite capabilities.

Chapter 2: Existing and Proposed System

2.1 Existing System

- Manual student enquiry handling.
- Information distributed across notices, documents, and departments.
- Repeated questions to college administration.

2.2 Proposed System

- Centralized Academic Information: A single digital repository for college data.
- Student Self-Service Portal: View profiles, timetables, and assignments.
- AI-based Question Answering: An offline chatbot that parses college-uploaded study materials to answer queries instantly.
- Administrative Information Management: A Next.js dashboard for admins.